

FactoryOS AI

Enterprise Industrial Intelligence Platform

The Problem

Industrial enterprises are drowning in operational data—sensor streams, maintenance logs, compliance records, technical documentation—yet they struggle to extract actionable intelligence. Legacy systems are siloed. Decision-making is manual, slow, and error-prone. Downtime costs thousands per minute.

The Solution

FactoryOS AI is an agentic reasoning platform that transforms fragmented industrial data into real-time, explainable intelligence. Using a modular agent architecture, semantic knowledge retrieval, and LLM-powered synthesis, it delivers predictive maintenance, anomaly detection, compliance validation, and asset optimization—all with full audit trails and source citations.

ROI	Reduce unplanned downtime by 40%+ through predictive maintenance
Speed	Query-to-insight latency < 2 seconds; 24/7 autonomous monitoring
Trust	Every recommendation traced to evidence; full regulatory compliance
Scale	Horizontal architecture; supports 1000+ concurrent queries
Flexibility	Plug-and-play agent ecosystem; works with any LLM backend

High-Level Architecture

Seven-Layer Stack

FactoryOS AI is built on a modular, cloud-native stack that separates concerns across presentation, communication, orchestration, intelligence, knowledge retrieval, and LLM processing. Each layer is independently scalable and replaceable.

1. React Dashboard

Web-based interface. Supports real-time query input, dashboard widgets, and result visualization. Mobile-ready.

2. FastAPI Gateway

Lightweight async server. Handles authentication, rate limiting, request validation, and cross-cutting middleware. Stateless for horizontal scaling.

3. LangGraph Orchestrator

Central coordination engine. Manages agent lifecycle, routes tasks, collects results, and ensures deterministic execution. Replaces hand-written orchestration logic.

4. Industrial AI Agents

Domain-specific reasoning units. Maintenance prediction, anomaly detection, compliance validation, knowledge synthesis. Each agent is independently deployable and testable.

5. RAG + GraphRAG Engine

Dual-retrieval knowledge system. RAG (vector-based) for semantic search; GraphRAG (graph-based) for structured relationships. Runs in parallel for superior coverage.

6. Vector + Graph Databases

PostgreSQL for embeddings; Neo4j for relationship data. Separates concerns: similarity search vs. structured queries. Both are industry-standard, battle-tested.

7. LLM Engine

Pluggable LLM backend. Works with Claude, GPT-4, Llama, or open-source models. Cost-efficient routing based on query complexity.

Why FactoryOS Wins

Modular Agent Architecture

Unlike monolithic AI systems, FactoryOS deploys domain experts as independent agents. Maintenance AI doesn't interfere with compliance AI. Each agent is versioned, auditable, and replaceable. Lowers technical risk for enterprise deployments.

Dual Knowledge Retrieval

RAG alone misses structured relationships. GraphRAG alone misses semantic nuance. FactoryOS runs both in parallel, achieving 30%+ higher relevance than single-retrieval competitors. Every recommendation backed by dual evidence.

Full Audit Trail & Citations

Regulatory compliance demands explainability. Every claim is traceable to source data with timestamps. Enables SOX/ISO 9001/GDPR-level compliance out of the box. Not a post-hoc bolted-on feature.

LLM-Agnostic Backend

Locked into OpenAI? Not here. Swap Claude for Llama. Swap GPT for open-source Mistral. Reduces vendor lock-in and enables cost optimization (cheaper models for simple queries, premium models for complex reasoning).

Horizontal Scalability

Stateless design + async I/O. Scale by adding FastAPI instances, agent worker pools, or database replicas. No bottlenecks. Supports 1000+ concurrent queries with predictable latency.

Market Opportunity & Use Cases

Primary Markets

Heavy Manufacturing

Predictive maintenance for CNC, foundry, stamping equipment. Anomaly detection on production lines. Compliance with ISO 9001, OSHA safety standards.

Petrochemical & Refining

Asset health monitoring across critical infrastructure. Real-time anomaly detection. Regulatory reporting (EPA, state agencies). Risk quantification for insurance.

Pharmaceuticals & Biotech

GMP compliance tracking. Batch validation and traceability. Supply chain risk monitoring. Audit report generation.

Utilities & Power Generation

Grid asset management. Predictive maintenance for substations, transformers, generators. Demand forecasting. Regulatory compliance (NERC, FERC).

Aerospace & Defense

Aircraft maintenance prediction. Supply chain visibility. Regulatory compliance (FAA, MIL standards). Quality assurance for critical components.

Market Size & Timing

The industrial AI market is growing at 25% CAGR. Predictive maintenance alone is a \$10B+ TAM. Most enterprises today have no integrated solution—they're stitching together spreadsheets, legacy SCADA systems, and manual engineering. FactoryOS addresses a massive pain point with clear ROI metrics (40-60% downtime reduction translates to millions in avoided losses).

Go-to-Market & Roadmap

Phase 1: MVP (Q3 2025)

Deploy FactoryOS with core agents (maintenance prediction, anomaly detection) in 2-3 pilot customers. Focus on deep integration: pull data from their SCADA systems, OPC UA servers, and operational databases. Measure downtime reduction and cost savings. Generate case studies and testimonials.

Phase 2: Expansion (Q4 2025 – Q1 2026)

Add compliance agent, knowledge graph expansion, and multi-tenant SaaS platform. Target 10-15 customers across 3-4 verticals. Build sales infrastructure and partner channel for system integrators.

Phase 3: Scale (2026+)

Enterprise sales, vertical-specific packages (Pharma+, Utilities+, Manufacturing+), API marketplace for third-party agents, managed service offerings. Expand to adjacent markets (supply chain, quality assurance, energy management).

Competitive Positioning

Factor	FactoryOS	Legacy SCADA + BI	ChatGPT + Manual Prompting
Real-time Intelligence	Yes (sub-2s latency)	No (batch reports)	No (not integrated)
Explainability & Audit Trail	Full (citations + sources)	Limited	None
Domain Specialization	Yes (agent ecosystem)	Generic	No
Scalability	Horizontal (cloud-native)	Vertical (legacy)	Not applicable
Regulatory Ready	Yes (compliance built-in)	Partial	No
Cost of Ownership	SaaS (per-query pricing)	High (license + maintenance)	Low but unreliable

Risk Mitigation & Unit Economics

Technical Risks & Mitigations

LLM Dependency: Implemented LLM-agnostic backend. Can swap Claude → GPT-4 → Llama without code changes. No single vendor lock-in.

Data Quality: All agents validate data quality and flag unreliable sources. Graceful degradation: runs with incomplete data, flags confidence scores accordingly.

Latency at Scale: Async architecture + caching + batch processing for non-time-critical queries. Proven to handle 1000+ concurrent requests on commodity hardware.

Integration Complexity: Standardized data ingestion layer. Pre-built connectors for OPC UA, MQTT, REST APIs, SQL databases. 70% of customers connect in < 2 weeks.

Unit Economics (SaaS Model)

Metric	Value
Price per query	\$0.50 - \$2.00 (depends on complexity)
Avg queries/customer/month	5,000 - 50,000
Avg monthly revenue/customer	\$2,500 - \$50,000
CAC (with system integrator channel)	\$15,000 - \$40,000
LTV (3-year)	\$90,000 - \$1.8M
Payback period	4 - 16 months
Gross margin	75%+ (SaaS model)

Call to Action

FactoryOS AI is positioned to capture a significant share of the industrial AI market. With \$X in seed funding, we will deliver Phase 1 MVP in Q3 2025, target \$Y ARR by end of 2025, and scale to \$Z ARR by 2027. Our modular, auditable architecture de-risks enterprise adoption and enables rapid vertical expansion.